Assignment- String

Assignment -1: Write a PHP program to:
Store your full name in a variable.
Display the length of the string using strlen().
Convert the name into uppercase and lowercase using strtoupper() and strtolower().
Reverse the name using strrev().

Code/Script:

PHP(Assignment1.php):

C: > wamp64 > www > String_130 > Assignment1.php

```php
<?php
// Assignment 1: Basic String Operations

$name = "Ayush Vyas";

// Length of the string
echo "Length of name: " . strlen(string: $name) . "<br>";

// Uppercase
echo "Uppercase: " . strtoupper(string: $name) . "<br>";

// Lowercase
echo "Lowercase: " . strtolower(string: $name) . "<br>";

// Reverse
echo "Reversed: " . strrev(string: $name) . "<br>";
?>
```

Output:

Assignment1:

localhost/String_130/Assignment1.php

Length of name: 10
Uppercase: AYUSH VYAS
Lowercase: ayush vyas
Reversed: sayV hsuyA

Assignment -2: Create a PHP program that:

Stores a sentence: “PHP is a popular general-purpose scripting language.”

Find the position of the word “general-purpose” using strpos().

Replace “PHP” with “Python” using str_replace().

Display the modified string.

Code/Script:

PHP(Assignment2.php):

C: > wamp64 > www > String_130 > Assignment2.php

```php
<?php
// Assignment 2: String Searching

$sentence = "PHP is a popular general-purpose scripting language.";

// Position of "general-purpose"
$pos = strpos(haystack: $sentence, needle: "general-purpose");
echo "Position of 'general-purpose': $pos<br>";

// Replace "PHP" with "Python"
$modified = str_replace(search: "PHP", replace: "Python", subject: $sentence);
echo "Modified String: $modified<br>";
?>
```

Output:

Assignment2:

localhost/String_130/Assignment2.php

Position of 'general-purpose': 17
Modified String: Python is a popular general-purpose scripting language.

Assignment -3: Write a PHP script that:

o Stores a string: “Welcome to University of Technology”

o Extract “University” using substr().

o Extract the last 10 characters of the string.

o Display only the first 5 characters.

Code/Script:

PHP(Assignment3.php):

C: > wamp64 > www > String_130 > Assignment3.php

```php
<?php
// Assignment 3: Substring Extraction

$str = "Welcome to University of Technology";

// Extract "University"
$university = substr(string: $str, offset: 11, length: 10);
echo "Extracted Word: $university<br>";

// Last 10 characters
$last10 = substr(string: $str, offset: -10);
echo "Last 10 Characters: $last10<br>";

// First 5 characters
$first5 = substr(string: $str, offset: 0, length: 5);
echo "First 5 Characters: $first5<br>";
?>
```

Output:

Assignment3:

localhost/String_130/Assignment3.php

Extracted Word: University
Last 10 Characters: Technology
First 5 Characters: Welco

Assignment -4: Write a PHP function to check whether a given word is a palindrome (same forward and backward).

Example Input: “level”, “madam”, “hello”

Expected Output:

level → Palindrome

madam → Palindrome

hello → Not a Palindrome

Code/Script:

PHP(Assignment4.php):

C: > wamp64 > www > String_130 > Assignment4.php > ...

```php
<?php
// Assignment 4: Palindrome Checker

function isPalindrome($word): bool {
    $word = strtolower(string: $word);
    return $word === strrev(string: $word);
}

$words = ["level", "madam", "hello"];

foreach ($words as $w) {
    if (isPalindrome(word: $w)) {
        echo "$w → Palindrome<br>";
    } else {
        echo "$w → Not a Palindrome<br>";
    }
}
?>
```

Output:

Assignment4:

localhost/String_130/Assignment4.php

level → Palindrome
madam → Palindrome
hello → Not a Palindrome

Assignment -5: Write a PHP program to:

Take a paragraph of text as input.

Count the number of words using str_word_count().

Count the number of occurrences of the word “PHP” (case-insensitive).

Display the most repeated word in the paragraph.

Code/Script:

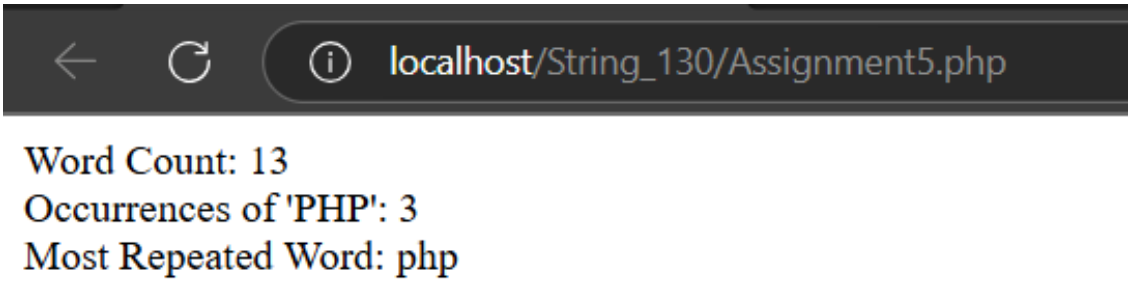
PHP(Assignment5.php):

C: > wamp64 > www > String_130 > Assignment5.php

```php
<?php
// Assignment 5: Word Counter

$paragraph = "PHP is powerful. PHP is popular. Many developers love PHP for web development.";

// Count words
echo "Word Count: " . str_word_count(string: $paragraph) . "<br>";

// Count occurrences of "PHP" (case-insensitive)
$countPHP = substr_count(haystack: strtolower(string: $paragraph), needle: "php");
echo "Occurrences of 'PHP': $countPHP<br>";

// Find most repeated word
$words = str_word_count(string: strtolower(string: $paragraph), format: 1);
$frequency = array_count_values(array: $words);
arsort(array: &$frequency);
$mostRepeated = array_key_first(array: $frequency);

echo "Most Repeated Word: $mostRepeated<br>";
?>
```

Output:

Assignment5:



Assignment -6: Write a PHP script that:

o Takes a CSV string: “apple,banana,grape,mango,orange”.

o Split it into an array using explode().

o Sort the array alphabetically.

o Join back into a string with “ | “ as a separator using implode().

Code/Script:

PHP(Assignment6.php):

C: > wamp64 > www > String_130 > Assignment6.php

```php
<?php
// Assignment 6: String Splitting & Joining

$csv = "apple,banana,grape,mango,orange";

// Split
$fruits = explode(separator: ",", string: $csv);

// Sort alphabetically
sort(array: &$fruits);

// Join with " | "
$joined = implode(separator: " | ", array: $fruits);

echo "Sorted & Joined String: $joined<br>";
?>
```

Output:

Assignment6:

localhost/String_130/Assignment6.php

Sorted & Joined String: apple | banana | grape | mango | orange

Assignment -7: Create a PHP function to validate a password string based on the following rules:

At least 8 characters long

Must contain at least one uppercase letter

Must contain at least one lowercase letter

Must contain at least one number

Must contain at least one special character (@, #, $, %, & etc.)

Return “Valid Password” or & “Invalid Password”

Code/Script:

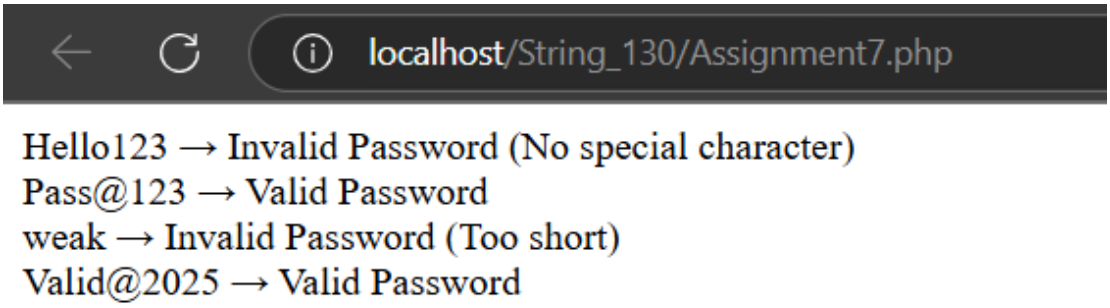
PHP(Assignment7.php):

C: > wamp64 > www > String_130 > Assignment7.php > ...

```php
<?php
// Assignment 7: Password Validation

1 reference
function validatePassword($password): string {
    if (strlen(string: $password) < 8) return "Invalid Password (Too short)";
    if (!preg_match(pattern: "/[A-Z]/", subject: $password)) return "Invalid Password (No uppercase letter)";
    if (!preg_match(pattern: "/[a-z]/", subject: $password)) return "Invalid Password (No lowercase letter)";
    if (!preg_match(pattern: "/[0-9]/", subject: $password)) return "Invalid Password (No number)";
    if (!preg_match(pattern: "/[@#$%&*!]/", subject: $password)) return "Invalid Password (No special character)";

    return "Valid Password";
}

// Test passwords
$passwords = ["Hello123", "Pass@123", "weak", "Valid@2025"];

foreach ($passwords as $pwd) {
    echo "$pwd → " . validatePassword(password: $pwd) . "<br>";
}
?>
```

Output:

Assignment7:



Conclusion / Learning Outcome:

Learning Outcomes:
Learned how to manipulate and analyse strings in PHP using built-in functions (strlen, substr, strpos, strrev, etc.).
Understood basic text processing techniques like searching, replacing, counting, and splitting strings in a procedural style.
Conclusion:
The procedural approach makes string handling simple and beginner friendly. By mastering PHP's built-in string functions, one can perform common operations like searching, extracting, validating, and analysing text effectively. This method is quick, readable, and best suited for small-scale projects.